

Typesetting Adlam Mathematical Equations in LibreOffice

Overview

Adlam is the writing system used primarily for the Fula language. The script runs completely right-to-left, unlike Arabic where numbers are composed left-to-right. Proper typesetting of Adlam mathematical equations requires a font that supports mirroring of mathematical symbols like XITS Math as well as a type setting program that supports right-to-left formula composition like LibreOffice.

Mathematical Conventions

	Example	Latin	(Adlam) Ɖɓɓɓ
Digits		1 2 3 4 5 6 7 8 9 0	Ɖ Ɛ 8 8 8 8 8 8 8 8 8 8
Letters		a-z/A-Z	Ɛ-ɓ\Ɛ-ɓ
Parameter/ Coefficients	$ax + by$	a-d	ɓ-ɓ
Cartesian Coordinates		x, y, z	ɓ ,3 ,ɓ
Length/Width		l/w	3\m
Radius/Diameter	$d = 2r$	r/d	ɓ\ɓ
Angles	$\cos(\theta)$	$\alpha, \beta, \gamma, \theta, \varphi$	ɓ ,θ ,ɓ ,θ ,ɓ
Functions	$f(x)$	f, g, h	ɓ ,ɓ ,ɓ
Decimal Separator	1.25	.	.
Thousands Separator	1,250	,	,
Mean		μ	ɓ
Integer Variables/Indices	$\sum_{i=1}^n i$	i-k	ɓ-ɓ
Fixed Integer/Count/ Probability	$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$	n	2

Typesetting

LibreOffice supports the mirroring of all mathematical symbols. This handles the majority of cases as long as the formula editor is set to Right-to-Left mode. This is done by looking for the following button.



For example the simply by placing the editor in Right-To-Left mode the following formulas automatically mirror.

LTR	RTL
$\square < \square$	$\square > \square$
$\square \leq \square$	$\square \geq \square$
$\sqrt{\square}$	$\sqrt{\square}$
$\square \in \square$	$\square \ni \square$
$\int \square$	$\square \int$
$\oint \square$	$\square \oint$

Functions with specific names have been added to the user interface. Below is a complete list of those that have been included.

Logarithms/Exponentials

Latin	Adlam
$\ln(\square)$	$(\square)_{\text{نل}}$
$\exp(\square)$	$(\square)_{\text{اڤ}}$
$\log(\square)$	$(\square)_{\text{لو}}$

Limits

Latin	Adlam
$\lim_{x \rightarrow a} f(x)$	ⵍⵉⵎ ⵏ ⵑⵓⵔ ⵏ ⵑⵓⵔ ⵏ ⵑⵓⵔ
$\liminf_{x \rightarrow a} f(x)$	ⵍⵉⵎ ⵏⵉⵎ ⵏ ⵑⵓⵔ ⵏ ⵑⵓⵔ
$\limsup_{x \rightarrow a} f(x)$	ⵍⵉⵎ ⵏⵉⵎ ⵏ ⵑⵓⵔ ⵏ ⵑⵓⵔ

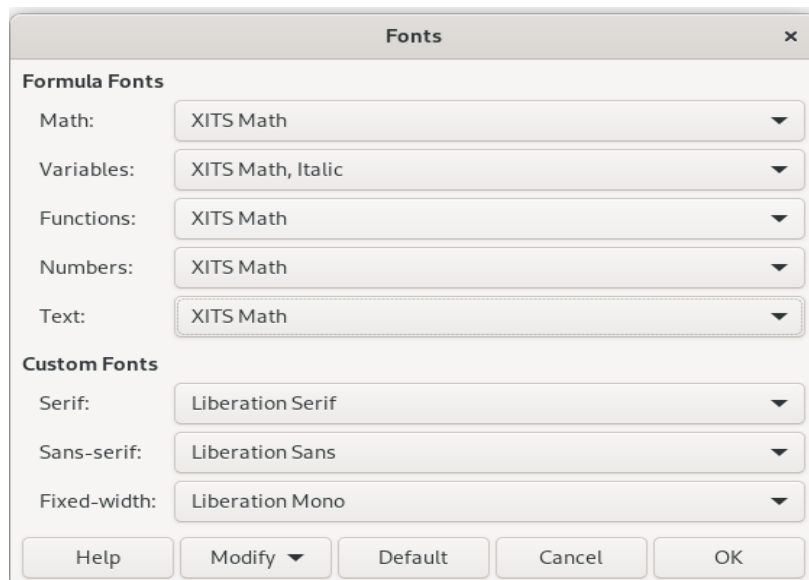
Trigonometry

Latin	Adlam
$\sin(x)$	(ⵑⵓⵔ) ⵉⵙⵉⵏ
$\cos(x)$	(ⵑⵓⵔ) ⵉⵙⵓⵏ
$\tan(x)$	(ⵑⵓⵔ) ⵉⵙⵓⵏ
$\sec(x)$	(ⵑⵓⵔ) ⵉⵙⵓⵏ
$\csc(x)$	(ⵑⵓⵔ) ⵉⵙⵓⵏ
$\cot(x)$	(ⵑⵓⵔ) ⵉⵙⵓⵏ
$\sinh(x)$	(ⵑⵓⵔ) ⵉⵙⵓⵏ
$\cosh(x)$	(ⵑⵓⵔ) ⵉⵙⵓⵏ
$\tanh(x)$	(ⵑⵓⵔ) ⵉⵙⵓⵏ
$\coth(x)$	(ⵑⵓⵔ) ⵉⵙⵓⵏ
$\operatorname{sech}(x)$	(ⵑⵓⵔ) ⵉⵙⵓⵏ
$\operatorname{csch}(x)$	(ⵑⵓⵔ) ⵉⵙⵓⵏ
$\arcsin(x)$	(ⵑⵓⵔ) ⵉⵙⵓⵏ
$\arccos(x)$	(ⵑⵓⵔ) ⵉⵙⵓⵏ
$\arctan(x)$	(ⵑⵓⵔ) ⵉⵙⵓⵏ

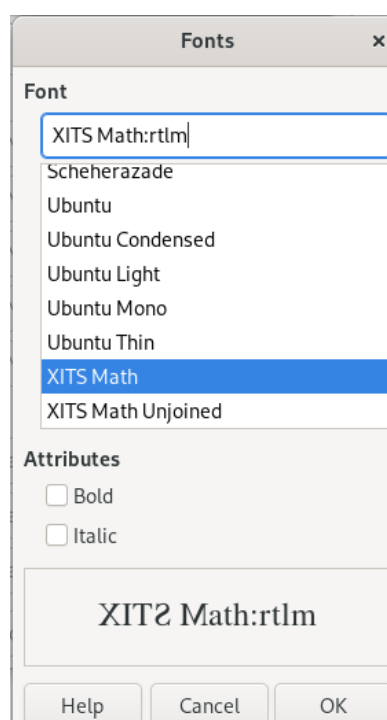
Font Choices

XITS Math includes 5 fonts: XITS Math Regular, XITS Math Bold, XITS Math Italic, XITS Math Unjoined Regular, and XITS Math Unjoined Italic. For typesetting Adlam only the first three fonts are needed with the bold being optional.

LibreOffice provides the ability to select the font for different elements of a formula. Below is an optimal setup. The bold font not necessary for legible equations but can be used if needed.



Adlam does use some non-standard mirroring characters for some formulas. To enable these enter “:rtlm” after the font name when selecting it.



Sample Formulas

Quadratic Equation:

$$\frac{bx \pm \sqrt{b^2 - 4ac}}{2a} = x$$

Slope Formula:

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}$$

Taylor Series:

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \dots$$

L'Hôpital's Rule:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

L'Hôpital's Rule with Trigonometry:

$$\lim_{x \rightarrow a} \frac{\sin(x)}{x} = \lim_{x \rightarrow a} \frac{\cos(x)}{1} = \cos(a)$$